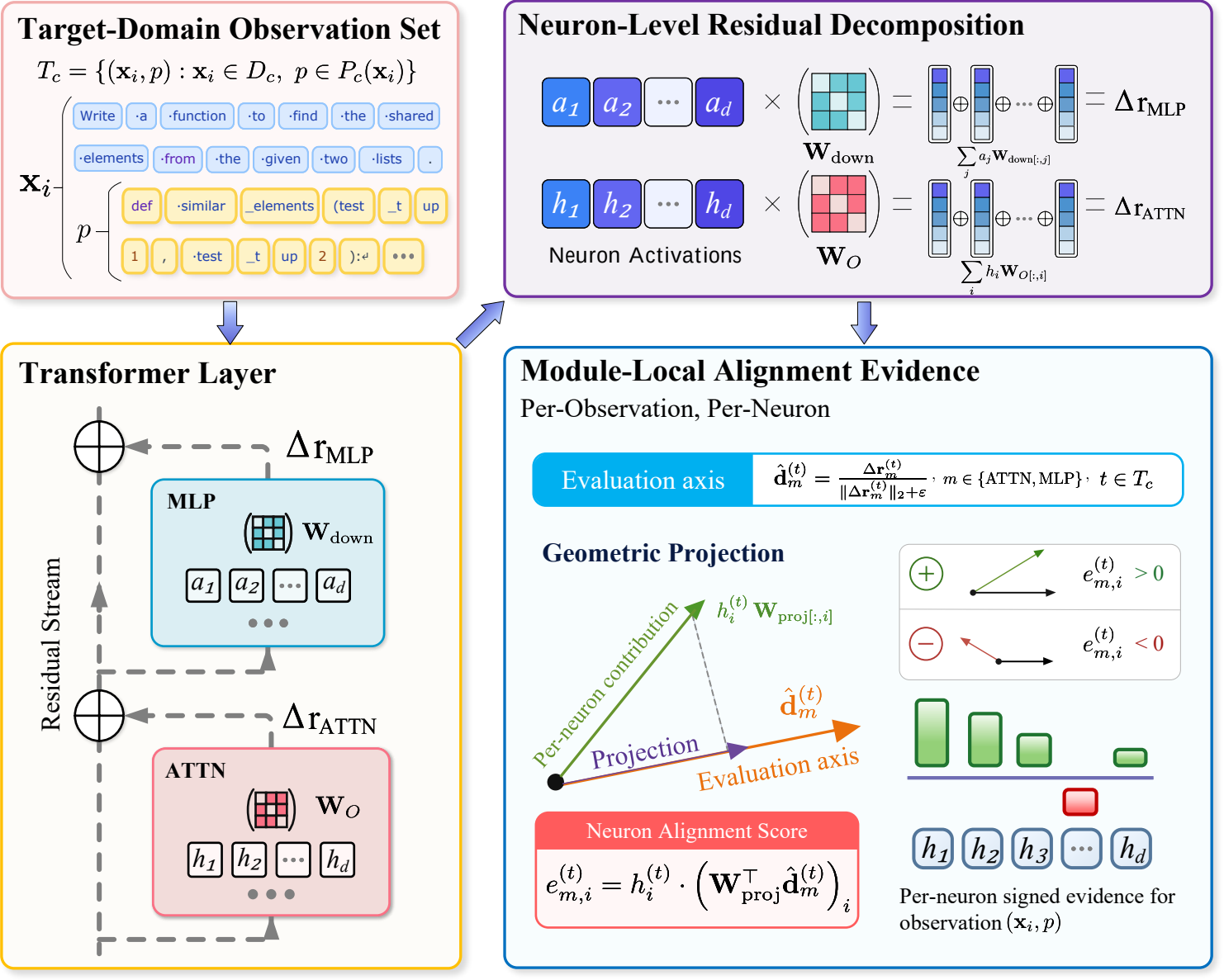

Target-Domain Observation Set
$T_c = \{(\mathbf{x}_i, p) : \mathbf{x}_i \in D_c,\ p \in P_c(\mathbf{x}_i)\}$
$\mathbf{x}_i$
Write ·a ·function ·to ·find ·the ·shared
·elements ·from ·the ·given ·two ·lists .
p
def ·similar _elements (test _t up
1 , ·test _t up 2):↵ •••
Neuron-Level Residual Decomposition
a_1 a_2 ⋯ a_d × $\mathbf{W}_{\text{down}}$ = $\sum_j a_j \mathbf{W}_{\text{down}[:,j]}$ = $\Delta \mathbf{r}_{\text{MLP}}$
h_1 h_2 ⋯ h_d × $\mathbf{W}_O$ = $\sum_i h_i \mathbf{W}_{O[:,i]}$ = $\Delta \mathbf{r}_{\text{ATTN}}$
Neuron Activations
Transformer Layer
Residual Stream
$\Delta \mathbf{r}_{\text{MLP}}$
MLP
$\mathbf{W}_{\text{down}}$
a_1 a_2 ⋯ a_d
$\Delta \mathbf{r}_{\text{ATTN}}$
ATTN
$\mathbf{W}_O$
h_1 h_2 ⋯ h_d
Module-Local Alignment Evidence
Per-Observation, Per-Neuron
Evaluation axis
$\hat{\mathbf{d}}_m^{(t)} = \frac{\Delta \mathbf{r}_m^{(t)}}{\|\Delta \mathbf{r}_m^{(t)}\|_2 + \varepsilon},\ m \in \{\text{ATTN}, \text{MLP}\},\ t \in T_c$
Geometric Projection
$h_i^{(t)} \mathbf{W}_{\text{proj}[:,i]}$
Per-neuron contribution
Projection
$\hat{\mathbf{d}}_m^{(t)}$
Evaluation axis
Neuron Alignment Score
$e_{m,i}^{(t)} = h_i^{(t)} \cdot \left(\mathbf{W}_{\text{proj}}^\top \hat{\mathbf{d}}_m^{(t)}\right)_i$
$e_{m,i}^{(t)} > 0$
$e_{m,i}^{(t)} < 0$
h_1 h_2 h_3 ⋯ h_d
Per-neuron signed evidence for observation $(\mathbf{x}_i, p)$